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RESEARCH ARTICLE-CIVIL ENGINEERING

Predicting the Compressive Strength of Pervious Cement Concrete based on Fast Genetic Programming Method

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Abstract
The primary objective of this paper is to develop an appropriate predictive formula for the compressive strength of pervious concrete, which depends on its mixture. This will allow for the improvement of the proportioning procedure that considers both the target porosity and target compressive strength. To achieve this, an effective computational strategy is first constructed and investigated for the creation of simple and easily applicable symbolic regression functions within the Genetic Programming-based Symbolic Regression framework. Recent advancements in fast logical parallelism and model-based algorithms are also applied to perform calculations on a large quantity of examples, with the aim of finding the most suitable analytical solutions at a low computational cost. Next, to assess the effectiveness of this model in predicting the compressive strength of concrete in general, computations are carried out using the well-known Yeh's dataset on conventional concrete compressive strength. This dataset has been extensively studied using both "black-box" and "white-box" machine learning algorithms. The results reveal that more suitable formulas can be generated through this computational process, compared to several scenarios discussed in the literature. Furthermore, the model is extended to pervious concrete, based on the dataset of 164 samples of 28-day compressive strength collected from 14 different sources. The findings for pervious concrete exhibited high accuracy compared to the most effective black-box models and micromechanical/empirical models, with a coefficient of determination of approximately 0.9 for simple predictive equations, thereby supporting the effectiveness of the proposed approach.

Keywords Symbolic regression · Genetic programming · Compressive strength · Machine learning · Pervious concrete · Proportioning procedure

1 Introduction

Over the last 100 years, concrete has emerged as one of the most widely used construction materials. Due to highly nonhomogeneous mixture and time depend, determining the relationships between the composition and performance of both conventional and non-conventional concrete remains a significant challenge, particularly with respect to the relationship between mixture proportion and uniaxial compressive strength. To date, much empirical/theoretical researches have been conducted to establish simplified empirical non-linear relationships between the compressive strength (CS) of cement concrete and its mixture. The commonly used empirical models, such as Abram's law, which states that the strength of a concrete mix is inversely related to the mass ratio of water to cement, are often limited to specific mixtures for which the mathematical correlations were established. Because of the multi-scale characteristic of cement-based materials [1], some theoretical analyses in the framework of micromechanical approach have been developed to predict the CS based on material properties. Aside from the advantage of accurately describing physical phenomena, the multi-scale approximations [2–4] also have drawbacks, as they may lack detailed information or have uncertain values regarding the material properties, size, and shape of aggregate and the interlayer transition zone for different scales. Over the last two decades, various machine learning algorithms have been used in efforts to achieve more

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accurate and practical results. Nunez et al. (2021) [5] conducted a comprehensive review of the state-of-the-art in estimating the compressive strength of modern concrete mixtures using computational intelligence. They reviewed 153 relevant peer-reviewed publications and found that the compressive strength of different types of novel concretes was most commonly analyzed using machine learning (ML) algorithms such as artificial neural networks (ANN), support vector machines (SVM), tree-based ensembles, fuzzy logic, and genetic algorithms. The authors found that ANN and Extreme Gradient Boosting (XGB) [6], a type of tree-based ensemble, demonstrated superior accuracy. However, the high computational cost and "black-box" nature of these methods was acknowledged as a significant disadvantage. On the other hand, the accuracy of "white-box" statistical regression methods was found to be less accurate compared to most ML techniques, but the mathematical expressions they produce, particularly simple expressions, are highly useful and attractive due to their simplicity, ease of interpretation, and direct application to engineering [7–9].

Inspired by the attractive properties of "white-box" models, this paper focuses on an alternative machine learning model: symbolic regression (SR). SR is defined as a method for finding a suitable mathematical model to describe observed data [10]. Unlike other models that require specific knowledge of the data structure or function forms, SR operates under the assumption that a mathematical space exists containing possible mathematical arguments. As a result, SR is believed to be a more suitable approach to solving the prediction problem compared to traditional analytical and numerical optimization methods (e.g., conjugate gradient). Recent advancements in computer science have resulted in significant progress in the field of SR, particularly in material sciences [11, 12]. The results show that Genetic Programming-based Symbolic Regression methods (GPSR) offer significant potential for both accuracy and simplicity compared to other SR frameworks, such as Bayesian or AI Feynman. In fact, numerous publications have utilized GPSR to construct predictive functions for concrete strength [13–15]. Alongside the advantages and drawbacks of this approach (easy to apply but not highly effective), most studies do not address the question of whether the derived equations are optimally efficient in terms of both effectiveness and the complexity of their solutions.

Portland cement pervious concrete (PC) is a unique type of concrete that exhibits an interconnected pore structure and high porosity ranging from 15 to 35%. Pavements constructed from PC offer several advantages such as mitigating the risk of flooding, reducing traffic noise, controlling surface temperature, and increasing groundwater levels. However, there are major challenges associated with this low-cost solution, including clogging, durability, and optimal mixture proportion design [16]. The design of the PC mixture is more complex than that of conventional concrete because both strength and porosity need to be considered concurrently, unlike standard concrete where only compressive strength is the primary concern. To simplify the design process, a target value for total porosity is selected initially, and then the compressive strength is estimated based on its relationship with total porosity, as outlined in [17, 18]. Due to the importance of the relationship between strength and porosity in the design of pervious concrete, the development of this relationship has been a topic of both theoretical and experimental research. Vu et al. (2021) [19] proposed a micro-mechanical model that connects porosity, strength, and several geometric and mechanical properties of the material. This model has shown effectiveness compared to other simple experimental models, although its complexity and suboptimal statistical properties raise questions about practical applicability. However, the identification of a rigorous theoretical relationship between strength and porosity is valuable. From a predictive perspective, it is feasible to search for simpler analytical expressions that relate strength and porosity with additional input variables such as water/cement ratio and admixtures.

For the reasons, the primary focus of this paper is to propose a data-driven approach for constructing an appropriate predictive formula for the compressive strength of pervious concrete, which depends on its mixture. This approach will facilitate the enhancement of the proportioning procedure, taking into account both the desired porosity and target compressive strength. To execute these main objectives, the following steps will be undertaken:

- First, we will explore a suitable computational strategy for constructing straightforward and easily applicable symbolic regression functions within the Genetic Programming-based Symbolic Regression framework. Addressing the limitations identified in the previous analysis, we believe that extensive computation with a large number of solutions will yield the best results. Therefore, recent advancements in fast logical parallelism and model-based algorithms will be integrated into the genetic programming framework. Additionally, a Pareto front construction algorithm will be implemented to select the most reasonable solutions.
- Subsequently, to assess the effectiveness of this model in predicting the compressive strength of concrete in general, computations will be conducted using the well-established Yeh's dataset [20] for conventional concrete compressive strength. The selection of the most suitable algorithm and hyperparameters will also be carried out.
- Finally, the model will be employed to investigate pervious concrete materials based on a database constructed from various sources. Several outcomes related to material behavior and component design will be proposed.

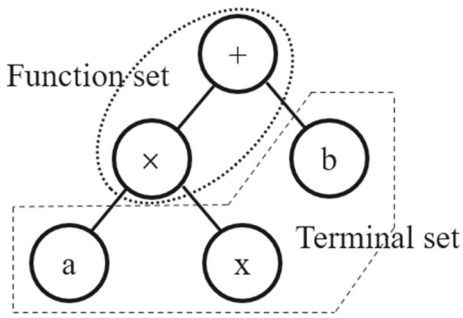

Fig. 1 Description of an individual in genetic programming: y = a × x + b

The paper is structured as follows: In Sect. 2, an overview of Genetic Programming fundamentals, encompassing fast logical parallelism, model-based algorithms, and a rational solution selection strategy, is provided. Moving forward to Sect. 3, a computational strategy is investigated, which is validated using a dataset of conventional concrete before its application to the pervious concrete dataset, comprising 164 samples of 28-day compressive strength collected from 14 distinct sources, as presented in Sect. 4. Ultimately, the concluding section synthesizes the principal discoveries of this research study.

2 Genetic Programming Framework

2.1 Basic of Genetic Programming

In the field of artificial intelligence, Genetic Programming (GP) is an automated approach that mimics natural selection to find an optimal result, first introduced by Koza in 1992 [21]. GP-based systems are based on the mechanisms of biological evolution and utilize an algorithm that employs random mutation, crossover, a fitness function, and multiple generations of evolution to find the optimal solution to user-defined problems. GP is a variation of the Genetic Algorithm (GA), with the main difference being the goal of the solution and the data structure used to represent an individual in a population. GA is often used to find the optimal solution to a problem, whereas GP is classified as a ML algorithm that finds the best model (such as a computer program or formula in symbolic regression problems) based on input data. Therefore, the model solution in GP has a tree-like data structure that consists of two main sets: the function set (including normal and logical operators) and the terminal set (composed of constants and input variables). Figure 1 illustrates an individual in GP, which represents the equation y = a × x + b.

In Genetic Programming (GP), the process of finding a suitable solution starts with an initial population. The individuals in the population are then evaluated through fitness conditions, where the fittest individuals are selected to survive and reproduce to form the next generation. This process of selection, crossover, mutation, and reproduction continues until a termination condition is met. The key components in GP are the definition of the initial population, the construction of the fitness function, the selection of genetic operators, and the determination of the termination conditions. Various GP methods have been developed to address different problems, and their details can be found in relevant literature in the field [22, 23]. Table 1 provides a practical description of the GP algorithm and its key parameters.

Table 1 Basic GP algorithm and key parameters

No	Process	Key parameter, input, output
1	Problem setting	Define input/output dataset Define function set, terminal set Define fitness function
2	Initial population setup Randomly initialize the initial population of formulas from the function set and the terminal set	Size of the population Maximum depth of the initial population Initialization method (Full/ Grow/ Ramped half-and-half, etc.)
3	Loop:	Number of generations
3.1	Calculate the fitness of the formulas in the population	The number of fitness cases
3.2	Select the most suitable formulas based on the fitness function	Tournament selection number
3.3	Create a new population based on the crossover and mutation operators	Crossover rate; Mutation rate
4	Until: a suitable formula is found or a specified stopping condition is reached	Satisfaction condition; Fitness better number; Runtime; Maximum number of generations
5	Finish Store the best possible equation	Result table

2.2 Fast Genetic Programming Models

Genetic programming (GP) is a popular and powerful optimization algorithm with a wide range of applications, including symbolic regression, classification, data mining… While implementing a genetic programming system may appear deceptively simple, selecting suitable solutions remains a time-consuming and challenging issue that limits the efficacy of GP in real-world applications. In the literature, many methods have been proposed to speed up Genetic Programming, which seem to correspond to the possibility of finding

better solutions [24]. Due to our background in civil engineering, this paper aims to leverage the most recent and effective approach to predict the compressive strength of concrete using symbolic regression modeling, rather than enhancing GP algorithms. Following the recommendation of Lim et al. (2014) [14], we have opted for the two most promising algorithms, identified by Operon [25] and GP-GOMEA [26], to carry out our computations. A brief summary of the algorithms is provided below, and more extensive information on them can be found in the accompanying documents.

The Genetic Gene-pool Optimal Mixing Evolutionary Algorithm (GP-GOMEA) is a model-based GP framework developed by Virgolin et al. (2021) [26]. Unlike traditional GP algorithms, which employ variation operators that act blindly, GP-GOMEA uses Gene-pool Optimal Mixing (GOM) algorithm to learn a model of interdependencies within the genotype, called the linkage model. This linkage model is then used to determine which patterns to propagate during variation. GP-GOMEA consists of two components: the learning of a linkage model and the application of the variation operator GOM to each solution in the population. The linkage model used in GP-GOMEA is the Family Of Subsets (FOS), which is a set of sets that contain indices representing positions in the genotype. Each set in the FOS specifies which parts of the genotype should be replaced in a bloc during variation. Once the FOS has been learned, the GOM operator generates the offspring population from the parent population and the FOS. Unlike standard crossover in GP, which swaps entire subtrees, GOM mixes potentially unconnected parts (i.e., tree nodes) of the fixed-size genotype. Additionally, instead of generating two new solutions from two parents, it creates an offspring by iteratively mixing a parent solution with multiple other solutions. The primary procedure of GP-GOMEA is depicted in Table 2.

Furthermore, GP-GOMEA employs the Linear Scaling technique to minimize the mean squared error of a GP tree. This is achieved by applying an optimal linear transformation to the output of the tree [27]. The primary idea behind LS is to enable GP to concentrate on the "shape" of the function to be approximated by providing translation and scaling coefficients that minimize the training Mean Squared Error.

Unlike standard tree-based GP algorithms, Operon [25] employs logical parallelism during evolution, generating each new offspring individual in its own logical thread. For each logical thread, Operon provides an offspring generator that enables the selection of two possible outcomes for a recombination event: success, when a suitable new offspring has been produced, and failure, when a new offspring cannot be produced. The main algorithm of Operon is depicted in the flowchart of Fig. 2.

In addition, Operon uses a linear encoding where each tree is represented as a postfix sequence of nodes. Each node has typical attributes such as length, depth, arity, or opcode. For example, the linear encoding for the formula $(a+b)\cdot(a+c)\cdot(b+c)$ corresponds to the postfix representation ((a b +) (a c +) (b c +) ×). Operon also introduces a new tree creation algorithm called the Balanced Tree Creator (BTC), which generates tree lengths from a given probability distribution and fine-tunes desired symbol frequencies in the population. The framework achieves efficient evaluation through the use of a batched tree interpreter, which iterates over the tree nodes and executes the corresponding functions on fixed-size batches of data. The fitness evaluation is performed by employing a local search step for optimal model parameters via non-linear least squares, provided by the Ceres library (http://ceres-solver.org/).

Table 2 Outline of GP-GOMEA and descriptions

No	Process	Description
1	**procedure** GP-GOMEA	
2	$\mathcal{P} \leftarrow IniPop\ (n^{pop})$	**Initial population setup** *IniPop:*Initialization method; $\mathcal{P}$: initial population; n^{pop}: population size
3	**while** *tm*() **do**	**Main loop of the model** *tm:* termination criterion (e.g., a limit on the number of generations or evaluations, or a maximum time)
4	$F \leftarrow learnFOS(\mathcal{P})$	**Create the linkage model** F: the linkage model (Family of Subsets)
5	$\mathcal{O} \leftarrow \varnothing$	**Loop of offspring operator** $\mathcal{O}$: offspring population Each solution $\mathcal{P}_i$ is used to generate an offspring solution $\mathcal{O}_i$ by GOM GOM: Gene-pool Optimal Mixing algorithm
6	**for** $i \in \{1, \ldots, n^{pop}\}$ **do** $\mathcal{O}_i \leftarrow GOM(\mathcal{P}, \mathcal{P}_i, F)\varnothing$	
7	$\mathcal{P} \leftarrow \mathcal{O}$	**Create new parent population** The offspring population $\mathcal{O}$ replace the parent population $\mathcal{P}$

2.3 Performance and Complexity Trade-off

In Nunez et al. (2021) [5], the accuracy indices of various computational intelligence models for estimating the compressive strength of modern concrete are compared. The coefficient of determination (denoted as R2 throughout the paper) is reported as the most frequently used evaluation metric in the literature. Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) are also commonly calculated. Hence, R2 is considered as the main accuracy metric, while

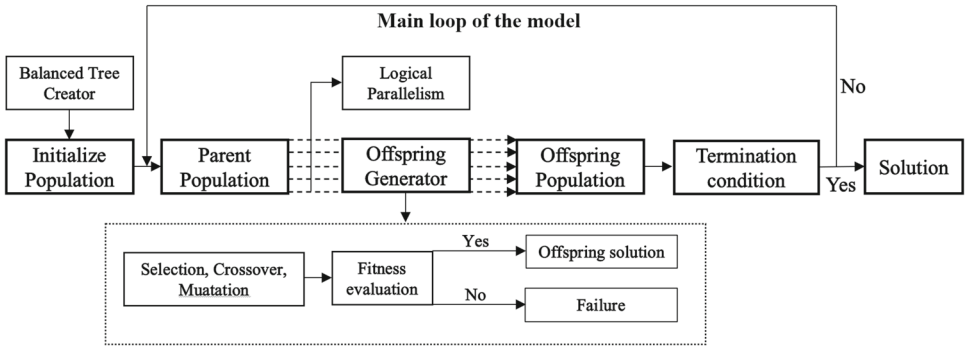

Fig. 2 Flowchart of the main algorithm of Operon

MAE and RMSE are still calculated and kept as important references. The definitions of R2, MAE, and RMSE can be found in numerous studies [28].

In the literature, the definition of a reasonable complexity of a mathematical expression is a topic of debate. In La Cava et al. (2021) [11], the complexity is defined as the number of mathematical operators, features, and constants in the model, also known as the node length in GP. For the purpose of consistency with the results obtained in La Cava et al. (2021) [11], this definition of complexity is adopted in this paper. However, this definition may not accurately reflect the interpretability of the models. Hence, selecting the best solutions that balance accuracy and complexity/interpretability is not a straightforward task. To address this challenge, a strategy that involves several steps is proposed: (i) conducting a large number of GP runs for each model; (ii) recording all relevant data (statistical measures for the training and testing datasets, equations obtained, complexity, GP parameters, run time, etc.) for each run of the model in a data file; (iii) constructing a Pareto front, which consists of optimal points that are not dominated by any other solutions in both accuracy and complexity; (iv) selecting the best optimal solutions directly from the solutions in the Pareto front.

3 Compressive Strength of Conventional Concrete

As mentioned earlier, the Genetic Programming-based Symbolic Regression (GPSR) method holds promise for investigating the strength properties of concrete and warrants further research. Before applying this approach to pervious concrete, we will assess the effectiveness of using the fast GPSR techniques described above on conventional concrete. The Yeh's dataset [20], a well-known dataset available online in the UC Irvine Machine Learning Repository (https://archive.ics.uci.edu/ml/datasets/concrete+compressive+strength), is used as a reference in the paper. It's worth noting that the UCI Machine Learning Repository, founded in 1987 by David Aha, a PhD student at the University of California, Irvine, encompasses a collection of databases, domain theories, and data generators. This repository serves as a crucial resource for the global machine learning community, offering access to approximately 644 datasets for empirical analysis in the field of machine learning. The statistical parameters of the dataset are presented in Table 3. The eight input variables, namely Cement, Blast Furnace Slag, Fly Ash, Water, Superplasticizer, Coarse Aggregate, Fine Aggregate, and Age, are denoted as X1 to X8, while Compressive Strength is denoted as Y for building symbolic regression equations. The "black-box" machine learning model based on the XGB algorithm achieved the best result with an R2 score of 0.93 on the test set [5]. In contrast, simple linear regression models were inadequate in producing suitable prediction functions for this dataset. Moreover, the Genetic Programming-based regression functions yielded the best results with a Mean Absolute Error (MAE) of approximately 6 MPa, as previously demonstrated by Mousavi et al. (2010, 2012) [29, 30].

Next, we employed the two fast GP models, namely Operon and GP-GOMEA, presented in Sect. 2, which allow for multiple feasible solutions within reasonable computational time, thereby increasing the likelihood of finding the best-fit solution. The basic GP model (referred to as gplearn [31]) was also computed and used as a reference. Based on previous studies and extensive trial tests, the impact of four primary parameters (population size: 100, 200… 1000; number of generations: 20, 40, 60, 80, 100; initial depth of population: 2, 3, 4, 5, 6; and tournament selection number: 5, 10, 15, 20) is evaluated on the results. For each run, the dataset is divided randomly into two groups: training (70% of the data) and testing (30% of the data). The GP algorithm randomly generated numerous functions, selected the best-fit symbolic regression formula using the training data, and then calculated statistical measures for the testing data. The calculation is repeated 10 times for each value of each parameter, resulting in a total of 10,000 calculations for each model. The results were recorded and saved for post-processing after each run. The main results were presented in the form of boxplots in Figs. 3, 4, and 5. The results indicated significant differences between the three GP models. In Fig. 3, the

Table 3 Statistical parameters of the Yeh's dataset

Unit	Cement kg/m^3	Blast furnace slag kg/m^3	Fly Ash kg/m^3	Water kg/m^3	Superplas-ticizer kg/m^3	Coarse aggregate kg/m^3	Fine aggregate kg/m^3	Age days	CS MPa
Notation	X1	X2	X3	X4	X5	X6	X7	X8	Y
mean	281.2	73.9	54.2	181.6	6.2	972.9	773.6	45.7	35.8
std	104.5	86.3	64.0	21.4	6.0	77.8	80.2	63.2	16.7
min	102.0	0.0	0.0	121.8	0.0	801.0	594.0	1.0	2.3
max	540.0	359.4	200.1	247.0	32.2	1145.0	992.6	365.0	82.6

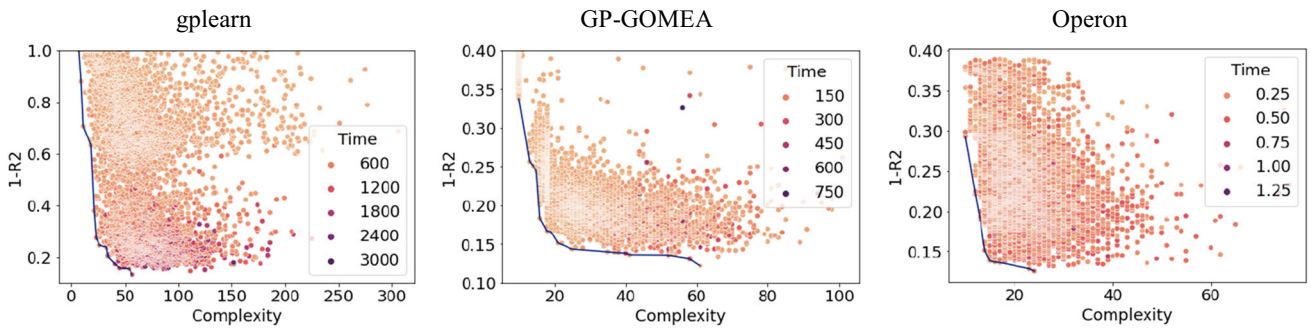

Fig. 3 (1-R2) – complexity – time calculations

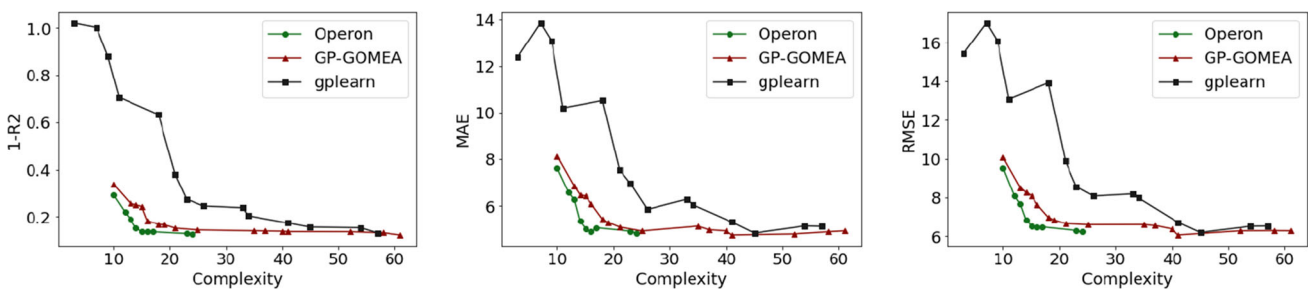

Fig. 4 Comparison of the trade-off accuracy complexity of the methods

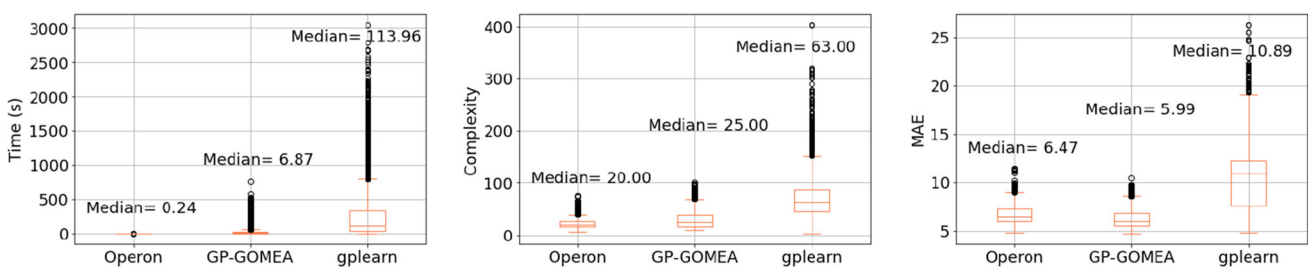

Fig. 5 The boxplots of time, complexity, and MAE of the methods

relationship between accuracy and complexity of the total solutions for the three methods is presented. Each point corresponded to a solution, with the run time (s) indicated by increasingly darker colors. The corresponding Pareto front (represented by the blue line) was also shown, consisting of optimal solutions that could not be improved in both accuracy and complexity. The accuracy was presented as (1-R2) in the plots to orient the Pareto front to the traditional position in the bottom left corner. The optimal solutions in the Pareto front and calculated the corresponding statistical parameters (MAE, RMSE) are also captured, and then compared the trade-off between accuracy and complexity of the methods in Fig. 4. Figure 5 displays boxplots depicting the time, complexity, and Mean Absolute Error (MAE) of the methods. Generally, we observed a significant gap between Operon and the other two methods in terms of the Pareto front and run

Table 4 Definition of the case's parameters

Parameter	Case 1	Case 2	Case 3
Size of the population	100, 200, 500, 700, 1000	100, 200, 500, 700, 1000, 1200, 1500, 1700, 2000	1000, 1001, 1002… 1998, 1999, 2000
Initial depth of the population	2, 3, 4, 5, 6	2, 3, 4… 10, 11, 12	5, 6, 7, 8
Number of generations	20, 40, 60, 80, 100	20, 40, 60… 180, 200	100, 101, 102… 198, 199, 200
Tournament selection number	5, 10, 15, 20	5, 10, 15… 35, 40	5, 10, 15, 20, 25

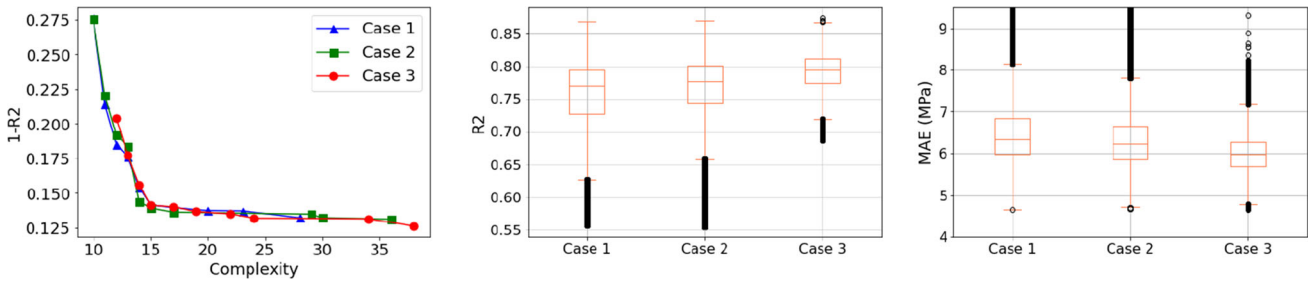

Fig. 6 Comparisons for the three Cases

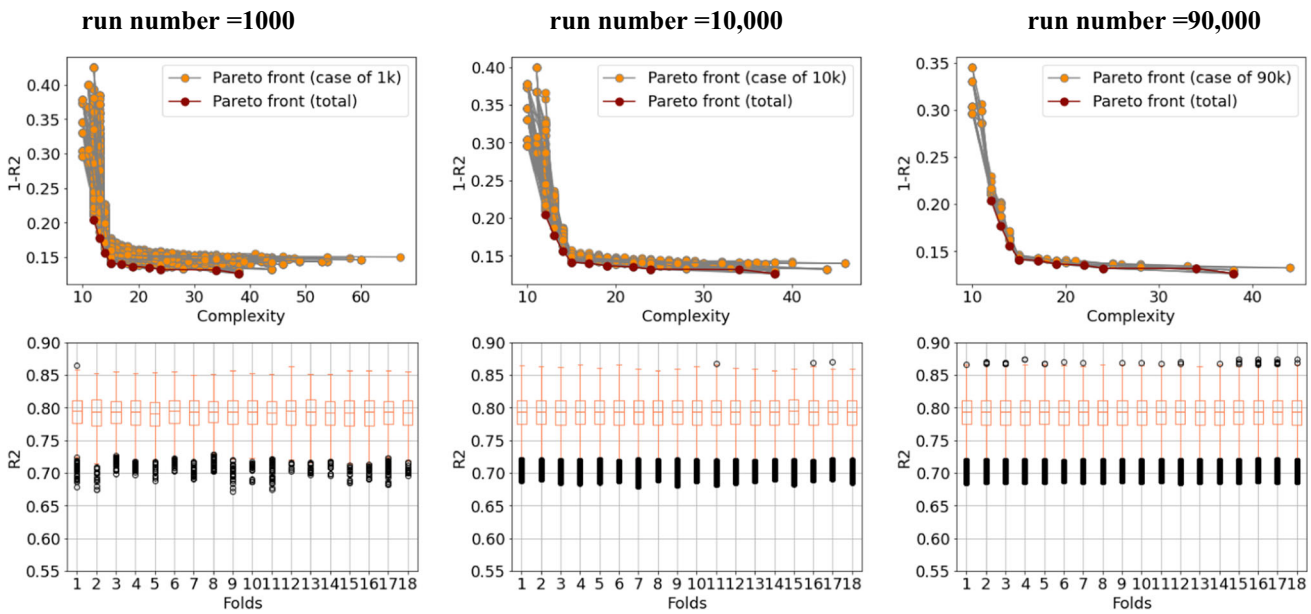

Fig. 7 Sensitivity of the run number for the Case 3

time. This makes Operon a more suitable choice for further studies on prediction problems of concrete properties based on structured data. In addition, the hardware used to run these tests was as follows: Intel Xeon E5 2689, 8 cores/16 threads, 2.6 GHz base frequency, 64 KB/256 KB/20 Mb L1/L2/L3 cache; 32 DDR3-1600 CL11 memory. All computations were programmed in Python basing on supports from the libraries such as Sklearn, Numpy, Pandas, Sympy.

In the second part of this section, the focus is on utilizing Operon for a more detailed examination. To identify the most impactful and important input parameters, three sets of calculations were conducted with a large number of runs (180,000 runs per set, totaling 540,000 runs for all) and various selections of the main parameters as presented in Table 4. In Fig. 6, comparisons are made between the three cases of calculation. A coherent relationship is observed between the

Pareto fronts generated from the three cases. It appears that the Pareto front is not influenced by the parameters of the GP model, as evidenced by the maximum accuracy values in Fig. 6, when a sufficient number of calculation runs are performed. However, the median accuracy in Case 3 is better than in Cases 1 and 2. This suggests that models with a greater diversity of parameter values are more effective in predicting results.

To determine the optimal number of calculation runs (referred to as the "run number"), a turning process was set up in Fig. 7 for three values of the run number: 1000, 10,000, and 90,000. For each scenario, 18 random folds were created from the total solutions of Case 3. In the first row of Fig. 7, the Pareto fronts of the folds corresponding to the three values of the run number are compared to the Pareto front of Case 3. The corresponding R2 values are presented in the second row. It is observed that the Pareto fronts converge when the run number reaches 90,000. However, the median value of R2 is relatively independent of the run number. For the purpose of searching for the best performing solutions, it is recommended to use a run number of 100,000 in combination with the GP parameters from Case 3 in further studies.

Based on the R2 values for the entire dataset, we have selected the solution with the highest accuracy, referred to as Formula 1, with R2 = 0.834, MAE = 5.24, and RMSE = 6.76 for the entire dataset.

$$Y = 65.24 \log \left(\frac{X4(\log(X4) + \log(X7) - \log(X8) + 0.242) + \sqrt{X3X8}\left(\sqrt{2}\log(X4) + 4.382\right) + (0.694(3.0X1 + 2.3X2))(\log(X8) + 1)}{X4} \right) - 177.34 \quad (1)$$

It is important to note that Formula 1 (1) yields statistical indices that are comparable to the analytical model proposed by Mousavi et al. (2012) [30] based on the Gene Express Programming method, with a slightly better balance between accuracy and complexity. Formula 1 is currently the best-performing formula for the Yeh dataset generated using GP techniques. However, it should be noted that these solutions are inflexible and tend to provide poor predictions for other independent datasets. Additionally, they are complex and lack theoretical meaning. Based on the solutions obtained at the Pareto fronts in Fig. 6, many solutions aim for a linear relationship between the product of the square of the logarithm of concrete age (T, in days) and the binder-water (B-W) mass ratio. Therefore, an adaptive linear regression model for uniaxial compressive strength (UCS) with two free parameters (a and b in MPa) is proposed that depends on the square of the logarithm of concrete age (T, in days) and the binder/water (B/W with B = X1 + X2 + X3, W = X4) mass ratio.

$$USC = a(\log T + 2)^{\frac{1}{2}} \times \left(\frac{B}{W}\right)^{\frac{1}{2}} - b \quad (2)$$

When applied to the entire dataset, Formula 2 yielded an R2 accuracy of 0.82. Additionally, the performance of the two proposed formulas using three datasets from the literature are evaluated. Dataset 1, consisting of 143 test results collected by Pala et al. (2004) [32], includes eight input variables, with Blast Furnace Slag being replaced by Silica Fume. Dataset 2, consisting of 200 samples, was collected by Videla et al. (2007) [33] and only includes Silica Fume as a cement replacement material. Dataset 3, collected by Lim et al. (2004) [14], includes the mass of Blast Furnace Slag as part of the cement mass and does not include Silica Fume or Fly Ash. The statistical analysis for Formula 2 indicates stability, with R2 values ranging from 0.838 for the Lim dataset (a = 29.64, b = 59.96), 0.389 for the Pala dataset, to 0.878 for the Videla dataset (a = 30.06, b = 38.71). However, Formula 1 was not suitable for predicting these datasets. This suggests that Formula 2 is flexible and appropriate for real-world engineering applications.

4 Compressive Strength of Pervious Concrete

In this section, we apply the analysis process described and qualified above to predict the compressive strength of previous concrete. Over the past few decades, Portland cement pervious concrete, also known as permeable concrete or porous cement concrete, has played an important role in sustainable urban drainage systems as a solution for increasing the permeability of surface layers by capturing rainfall in the base directly and efficiently. In general, pervious concrete is composed of quasi-uniform aggregates, cement, water, and potential admixtures, which results in a large, open pore structure (typically ranging from 15 to 35% by volume) and moderate strength capacity (typically ranging from 5 to 35 MPa). Pervious concrete has a simple mixture composition due to its low load-bearing requirements, while the impact of the pore network on the material properties plays a

Table 5 Statistics of the pervious concrete dataset (MS, AS, WC, AC, EP, CS stands for sand mass, minimum coarse aggregate size, water-to-cement ratio, aggregate-to cement ratio, effective porosity, compressive strength, respectively)

	Sand mass	Minimum coarse aggregate size	Water-to-cement ratio	Aggregate-to cement ratio	Effective porosity	Compressive strength
Unit	kg/m^3	mm	–	–	%	MPa
Notation	MS	AS	WC	AC	EP	CS
mean	56.48	5.25	0.34	4.73	22.47	17.07
std	105.10	2.83	0.07	1.86	8.42	10.32
min	0.00	1.18	0.21	2.24	3.68	1.06
max	340.00	12.50	0.55	12.00	42.00	49.00

key role. For example, common mixture design methods aim to achieve a desired porosity, and strength is estimated based on the porosity and mixture using several basic equations. Due to its simple composition and the natural relationship between porosity and strength, we believe that the SR method is useful for this type of material.

To demonstrate this, a dataset of 164 samples of 28-day compressive strength previous concrete, collected from 14 different sources [34–47] is used (see in Appendix). The three most important features affecting compressive strength in this dataset are the effective porosity, water-to-cement ratio, and aggregate size. Additionally, the aggregate-to-water ratio and mass of sand (fine aggregate) were also examined. Statistical parameters for the database are summarized in Table 5. The quality of the dataset is ensured by the large range of features, which covers all possible practical applications. The equivalent hardened compressive strength (at 28 days of hydration) and the effective porosity range from 1.06 to 49 MPa and 3.68 to 42%, respectively, covering typical ranges in practical applications (2.8 to 28 MPa for compressive strength and 15 to 35% for effective porosity, according to ACI 522R-10 [17]). The heatmap in Fig. 8 shows the correlation between features and between features and the target.

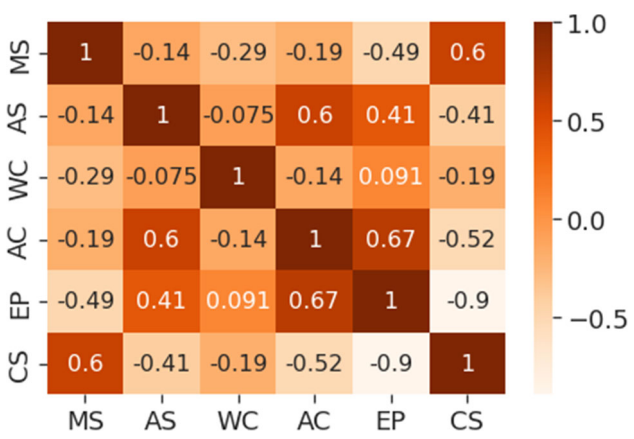

Fig. 8 The heatmap of the previous concrete dataset

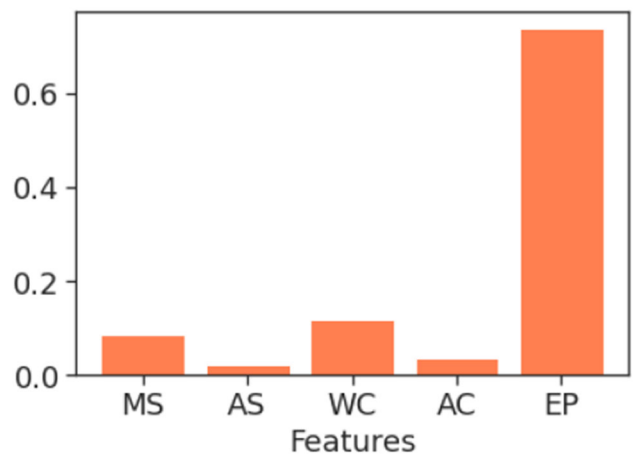

Fig. 9 Feature importance obtained by the XGB

The feature importance obtained through the XGB algorithm is displayed in Fig. 9. It can be observed that compressive strength and effective porosity have a close relationship, with compressive strength being inversely proportional to effective porosity and vice versa (correlation index of − 0.9).

In this section, a calculation procedure was performed to determine the most effective formula for predicting the compressive strength of pervious concrete. The results from the previous section indicate that the operon program with a greater range of parameter values appears to be the better selection (Case 3 in Table 4). In accordance with the procedure outlined above, after 100,000 runs of the algorithm, the main results are shown in Fig. 10. Figure 10a displays the accuracy (1-R2) and complexity of all results. The Pareto front, a line connecting the most useful candidate solutions, is also presented in Fig. 10a. Figure 10b shows box plots of the MAE and RMSE, demonstrating a slight grab of statistical parameters during the training and testing stages and indicating non-overfitting results for this approach. Finally, Fig. 10c counts the total number of inputs in the formulas. The AC input appeared in approximately 2% (2000 times) of the final formulas, while all other inputs were present in full in the final formulas.

Table 6 The proposed formulas for the pervious concrete dataset

Model	Description	Statistical indices
Formula 3	$189 - 113 \times \log(\mathrm{WC} + \log(\mathrm{AS} + \mathrm{EP}) + 1.0)$	R2 = 0.89, MAE = 2.53, RMSE = 3.38
Formula 4	$104.5 - 41.7\log\left(2.6\sqrt{(0.29 * \mathrm{AS} + \mathrm{WC} + 0.43 * \mathrm{EP})} - \mathrm{e}^{\left(-6.92\times 1.732^{6.0\mathrm{EP}}\times WC^{\left(3.0\mathrm{EP}+\frac{1}{2}\right)}\times \mathrm{AC}\times \mathrm{e}^{(-2.0\mathrm{MS})}\right)}\right)$	R2 = 0.915, MAE = 2.15, RMSE = 2.9
PSO-XGB	'learning_rate': 0.086, 'n_estimators': 872, 'reg_lambda': 4, 'max depth': 2	R2 = 0.915, MAE = 1.86, RMSE = 2.45

In Table 6, we present the two appropriate formulae from the solutions in the Pareto front of the full solutions. Formula 3 corresponds to the simplest equation with a complexity of 11, while Formula 4 corresponds to the most accurate solution. The mean statistical indices are summarized in Table 6 and are compared with the prediction of the "black box" PSO-XGB method for the testing stage. In fact, the PSO-XGB is structured from the XGB model, which is combined with the Particle Swarm Optimization (PSO) algorithm for optimizing nonlinear continuous functions. Details of the PSO-XGB method can be found in some recent publications [48]. It is necessary to note that the XGBoost (XGB) is actually one of the best regression machine learning methods for tabular data, as reported in Géron (2017) [49]. Significant results are obtained where the gap between the PSO-XGB and FORMULA 4 is very small, which is a remarkable result.

In Fig. 11, outcomes arising from Formula 3 for the total dataset of 164 samples are examined. To begin with, Fig. 11a depicts a strong correlation between the measured and predicted values, with an R2 of 0.89, MAE of 2.53 MPa, and RMSE of 3.38 MPa. An interesting aspect of the symbolic regression method is that it does not exhibit an overfitting effect when comparing the testing and training cases. As is well known, the compressive strength of pervious concrete is primarily influenced by its porosity, water-to-cement ratio, and aggregate size, as outlined in the theoretical framework for the fabrication of pervious concrete. In Vu et al. (2021) [19], a semi-empirical micromechanics-based formula was proposed by us, which fits the compressive strength-porosity relationship with a maximum possible R2 of 0.84, surpassing other traditional empirical laws such as logarithmic, power, and exponential equations. The Formula 3, which is a double logarithm of porosity, demonstrates better agreement with

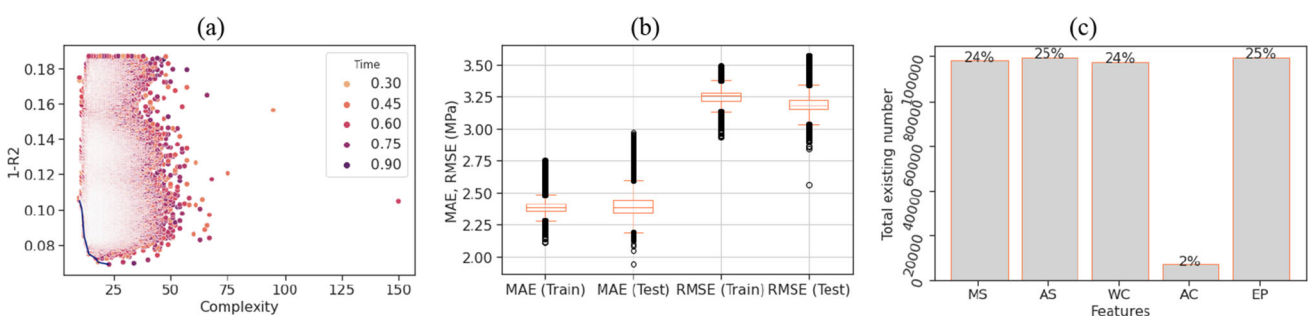

Fig. 10 Main results for the pervious concrete dataset

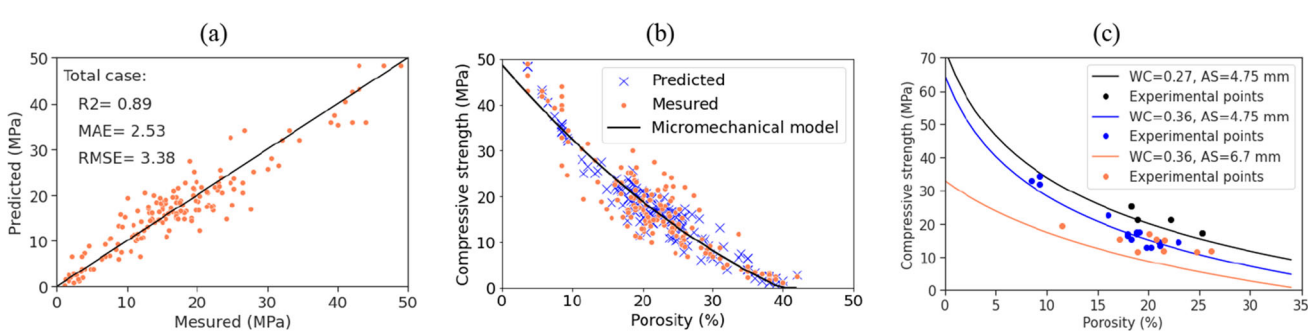

Fig. 11 Main results for the Formula 3

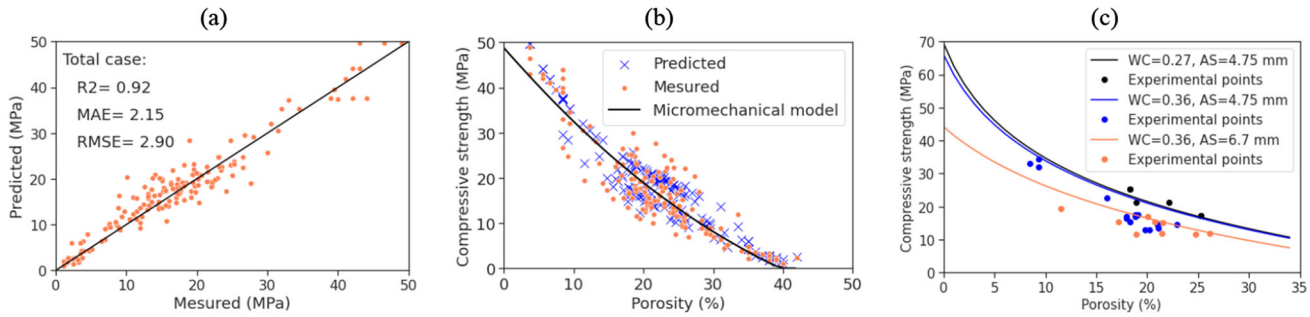

Fig. 12 Main results for the Formula 4

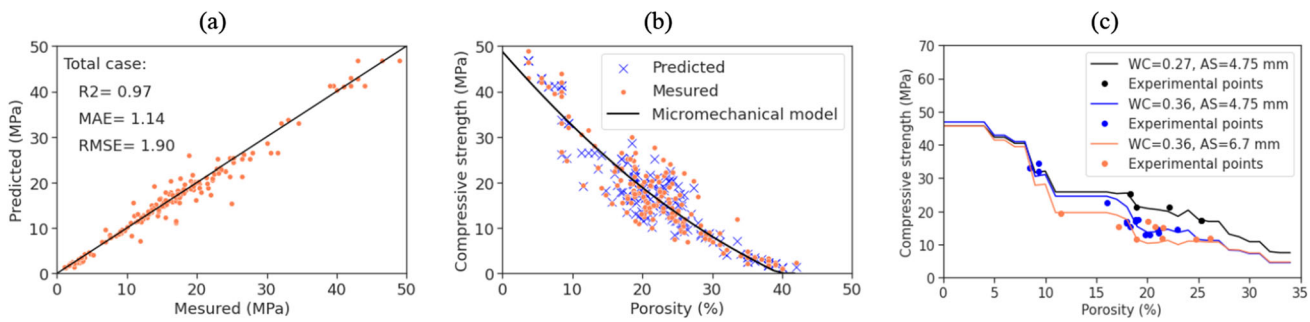

Fig. 13 Main results for the PSO-XGB

experimental data than the complex micromechanical model presented in Vu et al. (2021) [19]. The comparison is displayed in Fig. 11b. In Fig. 11c, Formula 3 is tested in three scenarios corresponding to WC = 0.27, AS = 4.75 mm; WC = 0.36, AS = 4.75 mm; and WC = 0.36, AS = 6.7 mm. The corresponding experimental points (solid circles of the same color as the theoretical lines) are also depicted in Fig. 11c.

Similar results for Formula 4 and PSO-XGB are presented in Figs. 12 and 13, respectively. Clearly, PSO-XGB achieved excellent predictions for the entire dataset, particularly in predicting the influence of water-cement ratio (WC) and aggregate sizes, as depicted in Fig. 13c. The difference between Formula 3 and Formula 4 is not significant, with the added note that Formula 4 does not clearly reflect the impact of WC on material behavior. Furthermore, Formula 3 demonstrates greater flexibility and suitability for normal pervious concrete mixture design, as it considers only the significant parameters of water-to-cement ratio and aggregate size. Specifically, compressive strength decreases (while porosity increases) as the WC ratio and/or the aggregate size increase, as depicted in Fig. 11c. This observation aligns with the principles outlined in the ACI 522R-10 report (Figs. 5.3 and 6.2 in ACI 522R-10).

To further validate our findings, we compared the predictive results obtained from Formula 3 with typical experimental data used for designing conventional pervious concrete mixtures according to ACI 522R-10 report, considering materials with component parameters of WC = 0.38 and aggregate sizes of No. 8 (AS = 2.35 mm) and No. 67 (AS = 4.75 mm). The comparison in Fig. 14 demonstrates a strong correlation between the predictive results and the independent dataset, with an R2 correlation coefficient of 0.96 for both aggregate sizes. The higher errors observed are primarily attributed to the larger porosity values, which is consistent with theoretical analyses suggesting increased sensitivity of results at higher porosity levels. This sensitivity is attributed to the greater influence of pore size and distribution, which Formula 3 does not explicitly account for.

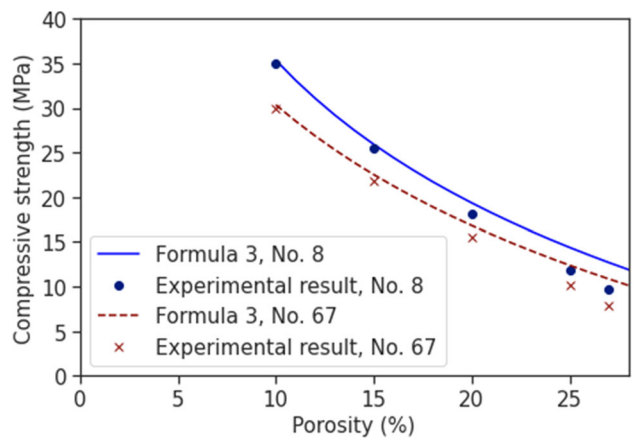

Fig. 14 Comparison of relationship compressive strength-porosity between the result from ACI 522R-10 and the prediction of Formula 3

Therefore, combining Formula 3 to enhance the proportioning procedure of pervious concrete mixture is feasible as it can simultaneously address both targets of porosity and

compressive strength. We propose the following improved proportioning procedure for conventional pervious concrete.

- Determine the target porosity and compressive strength. Typically, the porosity of pervious concrete ranges from 10 to 35%, and the compressive strength decreases from 35 to 5 MPa.
- Select aggregate sizes according to standard sieves and identify the minimum coarse aggregate size (AS).
- Calculate the water-cement (WC) ratio using the formula derived from Formula 3:

$$\mathrm{WC} = \exp((189 - \mathrm{CS})/113) - \log(\mathrm{AS} + \mathrm{EP}) - 1.0).$$

Note that the WC ratio for conventional pervious concrete typically varies between 0.27 and 0.45, as per ACI 522R-10.

- Select the paste volume V_p based on Fig. 6.3 in ACI 522R-10, which takes into account the influence of compaction.
- Determine the mass of cement, water, and aggregate:
  - o cement mass, $C = V_p/(\gamma_c^{-1} + WC) \times 1000$ (Kg), γ_c is specific gravity of cement,
  - o water mass, $W = WC \times C$ (Kg),
  - o aggregate mass, $A = (1 - V_p - EP) \times \gamma_a \times 1000$ (Kg), γ_a is specific gravity of aggregate.

Please note that this formula is constructed based on the dataset for conventional pervious concrete; hence, it may not be suitable for high-strength pervious concrete. Comparing it with recent studies on high-strength pervious concrete, it becomes evident that the increase in concrete strength results from an enhanced cement paste content. Therefore, it is sufficient to modify only two linear coefficients in Formula 3 (189 and 113) to achieve an approximation to experimental results within the typical porosity range of 15% to 25%. However, a strategic approach involving the construction of a sufficiently large dataset for high-strength pervious concrete is required to develop predictive formulas and suitable proportioning procedure.

5 Conclusions

In this study, we have introduced a data-driven approach that leverages optimal genetic programming techniques and data collection to explore mathematical expressions capable of predicting the compressive strength of both normal and pervious concrete. Our research has yielded the following key findings:

Firstly, the genetic programming process, particularly when coupled with the fast logical parallelism algorithm Operon, offers a more favorable balance between accuracy and simplicity, and demonstrates superior computational efficiency. This tool is highly valuable for engineering applications, especially when compared to graphical user interface (GUI) commercial software based on evolutionary or heuristic-based frameworks, such as Eureqa and Anova, which are also efficient but lack the ability to handle specific situations.

Secondly, in the context of conventional concrete, our findings have identified Formula 1 (Eq. 1) as the best-performing formula for the Yeh dataset generated using genetic programming techniques, while Formula 2 (Eq. 2) emerges as a simple and adaptable solution suitable for diverse datasets. In this regard, Formulas 1 and 2 stand out as among the most robust and practical empirical predictive equations for compressive strength of cement concrete based on its mixture. Their relevance for engineering applications is noteworthy and remarkable.

Thirdly, we extended our computational analysis to pervious concrete, using a dataset comprising 164 samples of 28-day compressive strength collected from 14 different sources. For the first time, we proposed two symbolic regression formulas to predict the compressive strength of pervious concrete. The more complex Formula 4 achieves the highest accuracy, with an R2 value of 0.915, on par with the accuracy of the best black box model (PSO-XGB). Meanwhile, the simpler logarithmic function-based Formula 3 provides a valuable tool for engineers when designing the composition of this material.

Fourthly, we employed Formula 3 to make qualitative and quantitative comparisons with several key outcomes outlined in the ACI 522R-10 report. Observable matches in the results were noted. Based on this, Formula 3 was integrated with the widely used ACI 522R-10 component design to propose a novel proportioning procedure that takes into consideration both target porosity and compressive strength.

Lastly, the application of the computational process developed here to other forms of non-conventional concrete materials can be considered. Furthermore, validation through experimental testing using the proposed results remains essential and is a future endeavor, even for pervious concrete.

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Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

The data for pervious concrete is available and attached to the paper.

No.	Sand mass	Minimum coarse aggregate size	Water-to-cement ratio	aggregate-to cement ratio	Effective porosity	compressive strength	Reference
Unit	kg/m^3	mm	–	–	%	MPa	
Notation	MS	AS	WC	AC	EP	CS	
1	340	4.75	0.3	3.35	20.44	14.1	[43]
2	340	4.75	0.3	3.63	21.42	15	
3	340	4.75	0.3	3.92	21.98	15	
4	340	4.75	0.3	4.06	22.23	16.3	
5	340	4.75	0.3	4.2	22.42	18.4	
6	340	4.75	0.3	4.48	22.51	18	
7	340	4.75	0.3	4.76	22.77	17.2	
8	0	4.75	0.33	4.99	34	3.9	[44]
9	0	4.75	0.33	4.99	31.5	4.5	
10	0	4.75	0.31	5.56	25.7	10.9	[45]
11	0	2.36	0.31	5.56	25.6	15.2	
12	0	2.36	0.31	5.56	24.7	16.5	
13	0	2.36	0.31	5.56	25.1	13.5	
14	0	4.75	0.31	5	25.1	13.9	
15	0	2.36	0.31	5	23.7	16.3	
16	0	2.36	0.31	5	23.1	17.9	
17	0	2.36	0.31	5	24.2	19.7	
18	0	4.75	0.31	4.55	23.9	19.1	
19	0	2.36	0.31	4.55	22.4	20.1	
20	0	2.36	0.31	4.55	23.1	21.6	
21	0	2.36	0.31	4.55	24.3	22.2	
22	0	4.75	0.31	4.17	22.5	18.1	
23	0	2.36	0.31	4.17	21.2	19.3	
24	0	2.36	0.31	4.17	20.5	21.5	
25	0	2.36	0.31	4.17	21.3	20.4	
26	0	4.75	0.31	3.85	20.3	14.7	
27	0	2.36	0.31	3.85	20.2	16.6	
28	0	2.36	0.31	3.85	19.9	19.5	
29	0	2.36	0.31	3.85	18.7	18.2	
30	0	2.36	0.31	4.55	23.3	22.2	
31	0	2.36	0.31	3.7	17.9	25.2	[47]
32	0	2.36	0.31	4	18.7	24.7	
33	0	2.36	0.31	4.35	21.1	19.4	
34	0	4.75	0.31	4.35	21.7	27.6	
35	0	4.75	0.31	4.76	24	24.1	
36	0	4.75	0.31	5.27	27.3	20.3	
37	0	4.8	0.35	3.7	18.64	19.73	[46]
38	0	4.8	0.35	2.88	9.2	24.72	
39	0	4.8	0.35	3.74	16.41	19.38	

No.	Sand mass	Minimum coarse aggregate size	Water-to-cement ratio	aggregate-to cement ratio	Effective porosity	compressive strength	Reference
Unit	kg/m^3	mm	–	–	%	MPa	
Notation	MS	AS	WC	AC	EP	CS	
40	0	4.8	0.35	2.91	8.45	26.73	
41	0	4.8	0.22	2.32	21.31	10.02	
42	0	6.3	0.35	3.31	14.8	17.3	[42]
43	0	8	0.35	3.3	15.6	15.3	
44	0	10	0.35	3.3	17.8	13.5	
45	0	12.5	0.35	3.3	19.5	11	
46	0	1.19	0.45	2.5	16.28	23.4	[41]
47	0	1.19	0.45	3	24.16	14.4	
48	0	1.19	0.45	3.5	28.37	10.6	
49	0	2.38	0.45	2.5	17.05	22.9	
50	0	2.38	0.45	3	24.56	12.9	
51	0	2.38	0.45	3.5	29.41	8.6	
52	0	4.75	0.45	2.5	20.4	18.1	
53	0	4.75	0.45	3	25.82	11.3	
54	0	4.75	0.45	3.5	29.57	9.1	
55	0	1.19	0.55	2.5	13.63	23.2	
56	0	1.19	0.55	3	23.88	12.4	
57	0	1.19	0.55	3.5	29.11	8.4	
58	0	2.38	0.55	2.5	18.75	17.6	
59	0	2.38	0.55	3	24.9	11.5	
60	0	2.38	0.55	3.5	29.59	7.5	
61	0	4.75	0.55	2.5	20.49	16	
62	0	4.75	0.55	3	25.89	10.5	
63	0	4.75	0.55	3.5	28.27	8.8	
64	0	4.5	0.35	8	35	2.21	[40]
65	0	9.5	0.35	8	38	2.45	
66	0	12.5	0.35	8	38	3.22	
67	0	4.5	0.35	8	42	2.42	
68	0	4.5	0.35	8	35	3.53	
69	0	9.5	0.35	8	38	2.8	
70	0	9.5	0.35	8	39	1.79	
71	0	9.5	0.35	8	35	3.92	
72	0	12.5	0.35	8	39	2.43	
73	0	12.5	0.35	8	36	3.67	
74	0	9.5	0.35	8	39	1.91	
75	0	9.5	0.35	8	35	3.59	
76	0	9.5	0.35	12	40	1.16	
77	0	12.5	0.35	12	40	1.06	
78	0	9.5	0.35	12	38	1.83	
79	0	9.5	0.35	7.2	35	4.76	
80	0	12.5	0.35	7.2	37	3.07	
81	0	9.5	0.35	7.2	36	3.45	
82	0	9.5	0.35	10	38	1.71	
83	0	12.5	0.35	10	38	1.67	

No.	Sand mass	Minimum coarse aggregate size	Water-to-cement ratio	aggregate-to cement ratio	Effective porosity	compressive strength	Reference
Unit	kg/m^3	mm	–	–	%	MPa	
Notation	MS	AS	WC	AC	EP	CS	
84	0	9.5	0.35	10	38	1.71	
85	0	9.5	0.35	6	30	6.95	
86	0	12.5	0.35	6	32	5.14	
87	0	9.5	0.35	6	31	6.45	
88	180	2.36	0.23	4.37	23	13.7	[39]
89	170	2.36	0.24	4.38	27	13.1	
90	160	2.36	0.23	4.39	31	8.8	
91	0	2.36	0.32	2.28	18.3	19.89	[38]
92	0	2.36	0.33	2.58	21.1	14.65	
93	0	2.36	0.33	3.13	25.5	9.66	
94	0	4.75	0.32	2.24	16.9	20.6	
95	0	4.75	0.33	2.61	22.3	15.89	
96	0	4.75	0.33	3.09	25.6	9.3	
97	0	9.5	0.32	2.39	19.5	19.13	
98	0	9.5	0.33	2.67	23.8	15.66	
99	0	9.5	0.33	3.28	24.6	7.55	
100	0	2.36	0.32	4.98	20.2	12.51	
101	0	4.75	0.32	4.67	19.5	17.61	
102	0	4.75	0.32	5.5	24.2	13.2	
103	0	4.75	0.33	7.65	28.9	7.7	
104	0	9.5	0.32	4.85	17.8	17.01	
105	0	9.5	0.32	5.74	24.2	12.24	
106	0	9.5	0.33	8.17	26.4	6.9	
107	107	1.18	0.26	3.74	26	22	[37]
108	109	1.18	0.36	3.69	23	25	
109	109	1.18	0.26	3.69	18.5	30	
110	100	2.36	0.25	4.14	28	17	
111	0	6.7	0.36	4.55	21.442	12	[36]
112	0	6.7	0.36	4.55	26.1396	12	
113	0	6.7	0.36	4.55	24.7444	11.5	
114	0	4.75	0.36	4.55	19.0868	17.5	
115	0	4.75	0.36	4.55	22.9524	14.5	
116	0	4.75	0.36	4.55	21.0196	14.5	
117	0	6.7	0.36	4.55	17.154	15.5	
118	0	6.7	0.36	4.55	11.5476	19.5	
119	0	6.7	0.36	4.55	18.9332	11.5	
120	0	4.75	0.36	4.55	18.3188	15.5	
121	0	4.75	0.36	4.55	20.1876	13	
122	0	4.75	0.36	4.55	21.122	13.5	
123	0	4.75	0.36	4.55	17.922	17	
124	0	4.75	0.36	4.55	17.922	16.5	
125	0	4.75	0.36	4.55	19.8164	13	
126	0	6.7	0.36	4.55	21.57	15	

No.	Sand mass	Minimum coarse aggregate size	Water-to-cement ratio	aggregate-to cement ratio	Effective porosity	compressive strength	Reference
Unit	kg/m^3	mm	–	–	%	MPa	
Notation	MS	AS	WC	AC	EP	CS	
127	0	6.7	0.36	4.55	20.0468	17	
128	0	6.7	0.36	4.55	20.7636	15.5	
129	0	4.75	0.36	4.55	18.8692	17	
130	0	4.75	0.36	4.55	16.0276	22.5	
131	0	4.75	0.36	4.55	18.8692	17.5	
132	242.41	4.75	0.38	4	11.2788	30.5	
133	240.07	4.75	0.38	4	12.226	31.5	
134	237.74	4.75	0.38	4	13.1732	28	
135	247.87	4.75	0.36	4	9.3716	34.5	
136	247.87	4.75	0.36	4	9.3716	32	
137	250.21	4.75	0.36	4	8.4244	33	
138	262.75	4.75	0.34	4	3.6756	49	
139	262.75	4.75	0.34	4	3.6756	46.5	
140	262.75	4.75	0.34	4	3.6756	43	
141	254.20	4.75	0.32	4	7.4644	39.5	
142	258.90	4.75	0.32	4	5.57	42	
143	251.84	4.75	0.32	4	8.4116	40	
144	257.39	4.75	0.3	4	6.5172	41	
145	257.39	4.75	0.3	4	6.5172	41	
146	252.67	4.75	0.3	4	8.4116	39	
147	252.67	4.75	0.3	4	8.4116	42	
148	252.67	4.75	0.3	4	8.4116	44	
149	259.75	4.75	0.3	4	5.57	43	
150	232.18	4.75	0.28	4	16.9492	23	
151	232.18	4.75	0.28	4	16.9492	26.5	
152	227.44	4.75	0.28	4	18.8436	23.5	
153	0	4.75	0.27	4.67	25.3	17.3	[35]
154	104	4.75	0.27	4.37	18.3	25.2	
155	0	4.75	0.27	4.5	25.3	17.3	[34]
156	0	9	0.27	4.5	33.6	11.9	
157	100	9	0.22	4.81	20.2	20.2	
158	100	4.75	0.27	4.38	18.3	25.2	
159	100	4.75	0.22	4.81	19	23.1	
160	100	4.75	0.21	4.61	26	9	
161	100	4.75	0.24	5.15	14.1	18.9	
162	0	4.75	0.27	4.5	18.9	21.4	
163	0	4.75	0.27	4.5	22.1	21.4	
164	100	4.75	0.27	4.38	19	26.5	

References

1. Constantinides, G.; Ulm, F.-J.: The effect of two types of CSH on the elasticity of cement-based materials: results from nanoindentation and micromechanical modeling. Cem. Concr. Res. **34**, 67–80 (2004)
2. Pichler, B.; Hellmich, C.: Upscaling quasi-brittle strength of cement paste and mortar: a multi-scale engineering mechanics model. Cem. Concr. Res. **41**, 467–476 (2011). https://doi.org/10.1016/j.cemconres.2011.01.010
3. Tran, B.-V.; Pham, D.-C.; Loc, M.-D.; Le, M.-C.: An adaptive approach for the chloride diffusivity of cement-based materials. Comput. Concr. **23**, 145–153 (2019). https://doi.org/10.12989/cac.2019.23.2.145
4. Tsivolas, E.; Gergidis, L.N.; Paipetis, A.S.: Multiscale modeling of extrinsic self healing GFRP materials. Modelling Simul. Mater. Sci. Eng. **31**, 025002 (2022). https://doi.org/10.1088/1361-651X/acaa49
5. Nunez, I.; Marani, A.; Flah, M.; Nehdi, M.L.: Estimating compressive strength of modern concrete mixtures using computational intelligence: a systematic review. Constr. Build. Mater. **310**, 125279 (2021). https://doi.org/10.1016/j.conbuildmat.2021.125279
6. Nguyen, H.; Nguyen, N.-M.; Cao, M.-T.; Hoang, N.-D.; Tran, X.-L.: Prediction of long-term deflections of reinforced-concrete members using a novel swarm optimized extreme gradient boosting machine. Eng. Comput. **38**, 1255–1267 (2022). https://doi.org/10.1007/s00366-020-01260-z
7. Behnood, A.; Golafshani, E.M.: Predicting the compressive strength of silica fume concrete using hybrid artificial neural network with multi-objective grey wolves. J. Clean. Prod. **202**, 54–64 (2018). https://doi.org/10.1016/j.jclepro.2018.08.065
8. Moghaddas, S.A.; Nekoei, M.; Mohammadi Golafshani, E.; Nehdi, M.; Arashpour, M.: Modeling carbonation depth of recycled aggregate concrete using novel automatic regression technique. J. Clean. Prod. **371**, 133522 (2022). https://doi.org/10.1016/j.jclepro.2022.133522
9. Özdemir, E.: A new predictive model for uniaxial compressive strength of rock using machine learning method: artificial intelligence-based age-layered population structure genetic programming (ALPS-GP). Arab. J. Sci. Eng. **47**, 629–639 (2022). https://doi.org/10.1007/s13369-021-05761-x
10. Duffy, J.; Engle-Warnick, J.: Using Symbolic Regression to Infer Strategies from Experimental Data. In: Chen, S.-H. (Ed.) Evolutionary Computation in Economics and Finance, pp. 61–82. Physica-Verlag HD, Heidelberg (2002)
11. La Cava, W.; Orzechowski, P.; Burlacu, B.; de França, F.O.; Virgolin, M.; Jin, Y.; Kommenda, M.; Moore, J.H.: Contemporary Symbolic Regression Methods and their Relative Performance, http://arxiv.org/abs/2107.14351, (2021)
12. Wang, Y.; Wagner, N.; Rondinelli, J.M.: Symbolic regression in materials science. MRS Commun. **9**, 793–805 (2019). https://doi.org/10.1557/mrc.2019.85
13. Chen, L.: Study of applying macroevolutionary genetic programming to concrete strength estimation. J. Comput. Civ. Eng. **17**, 290–294 (2003). https://doi.org/10.1061/(ASCE)0887-3801(2003)17:4(290)
14. Lim, C.-H.; Yoon, Y.-S.; Kim, J.-H.: Genetic algorithm in mix proportioning of high-performance concrete. Cem. Concr. Res. **34**, 409–420 (2004). https://doi.org/10.1016/j.cemconres.2003.08.018
15. Gholampour, A.; Gandomi, A.H.; Ozbakkaloglu, T.: New formulations for mechanical properties of recycled aggregate concrete using gene expression programming. Constr. Build. Mater. **130**, 122–145 (2017). https://doi.org/10.1016/j.conbuildmat.2016.10.114
16. Ferguson, B.: Porous Pavements. CRC Press (2005)
17. ACI Committee 522: 522R-10: Report on Pervious Concrete. Technical Documents. (2010)
18. Xie, X.; Zhang, T.; Wang, C.; Yang, Y.; Bogush, A.; Khayrulina, E.; Huang, Z.; Wei, J.; Yu, Q.: Mixture proportion design of pervious concrete based on the relationships between fundamental properties and skeleton structures. Cement Concr. Compos. **113**, 103693 (2020). https://doi.org/10.1016/j.cemconcomp.2020.103693
19. Vu, V.-H.; Tran, B.-V.; Le, B.-A.; Nguyen, H.-Q.: Prediction of the relationship between strength and porosity of pervious concrete: a micromechanical investigation. Mech. Res. Commun. **118**, 103791 (2021). https://doi.org/10.1016/j.mechrescom.2021.103791
20. Yeh, I.-C.: Modeling of strength of high-performance concrete using artificial neural networks. Cem. Concr. Res. **28**, 1797–1808 (1998). https://doi.org/10.1016/S0008-8846(98)00165-3
21. Koza, J.R.: Genetic Programming: On the Programming of Computers by Means of Natural Selection. Bradford Books, Cambridge, MA, USA (1992)
22. Poli, R.; Langdon, W.; Mcphee, N.: A Field Guide to Genetic Programming. Published via \texttt{http://lulu.com} and freely available at \texttt{http://www.gp-field-guide.org.uk (2008)
23. Affenzeller, M.; Wagner, S.; Winkler, S.; Beham, A.: Genetic algorithms and genetic programming: modern concepts and practical applications. CRC Press (2009)
24. Huang, Z.; Zhong, J.; Feng, L.; Mei, Y.; Cai, W.: A fast parallel genetic programming framework with adaptively weighted primitives for symbolic regression. Soft. Comput. **24**, 7523–7539 (2020). https://doi.org/10.1007/s00500-019-04379-4
25. Burlacu, B.; Kronberger, G.; Kommenda, M.: Operon C++: an efficient genetic programming framework for symbolic regression. In: Proceedings of the 2020 Genetic and Evolutionary Computation Conference Companion. pp. 1562–1570. Association for Computing Machinery, New York, NY, USA (2020)
26. Virgolin, M.; Alderliesten, T.; Witteveen, C.; Bosman, P.A.N.: Improving model-based genetic programming for symbolic regression of small expressions. Evol. Comput. **29**, 211–237 (2021). https://doi.org/10.1162/evco_a_00278
27. Keijzer, M.: Improving Symbolic Regression with Interval Arithmetic and Linear Scaling. In: Ryan, C.; Soule, T.; Keijzer, M.; Tsang, E.; Poli, R.; Costa, E. (Eds.) Genetic Programming, pp. 70–82. Springer, Berlin, Heidelberg (2003)
28. Nguyen-Sy, T.; Vu, M.-N.; Tran-Le, A.-D.; Tran, B.-V.; Nguyen, T.-T.-N.; Nguyen, T.-T.: Studying petrophysical properties of micritic limestones using machine learning methods. J. Appl. Geophys. (2020). https://doi.org/10.1016/j.jappgeo.2020.104226
29. Mousavi, S.M.; Gandomi, A.H.; Alavi, A.H.; Vesalimahmood, M.: Modeling of compressive strength of HPC mixes using a combined algorithm of genetic programming and orthogonal least squares. Struct. Eng. Mech. **36**, 225–241 (2010). https://doi.org/10.12989/sem.2010.36.2.225
30. Mousavi, S.M.; Aminian, P.; Gandomi, A.H.; Alavi, A.H.; Bolandi, H.: A new predictive model for compressive strength of HPC using gene expression programming. Adv. Eng. Softw. **45**, 105–114 (2012). https://doi.org/10.1016/j.advengsoft.2011.09.014
31. Trevor Stevens: gplearn Documention v0.4.1, (2022)
32. Pala, M.; Özbay, E.; Öztaş, A.; Yuce, M.I.: Appraisal of long-term effects of fly ash and silica fume on compressive strength of concrete by neural networks. Constr. Build. Mater. **21**, 384–394 (2007). https://doi.org/10.1016/j.conbuildmat.2005.08.009
33. Videla, C.; Gaedicke, C.: Modeling portland blast-furnace slag cement high-performance concrete. MJ. **101**, 365–375 (2004). https://doi.org/10.14359/13422
34. Wang, K.; Schaefer, V.; Kevern, J.: Development of Mix Proportion for Functional and Durable Pervious Concrete. In: Proceedings of the NRMCA concrete technology forum: Focus on Pervious Concrete 1–12, May 24–25, Nashville, TN, USA (2006)

35. Kevern, J.; Schaefer, V.; Wang, K.; Suleiman, M.: Pervious concrete mixture proportions for improved freeze-thaw durability. J. ASTM Int. **5**, 1–12 (2008). https://doi.org/10.1520/JAI101320
36. Lian, C.; Zhuge, Y.; Beecham, S.: The relationship between porosity and strength for porous concrete. Constr. Build. Mater. **25**, 4294–4298 (2011). https://doi.org/10.1016/j.conbuildmat.2011.05.005
37. Shu, X.; Huang, B.; Wu, H.; Dong, Q.; Burdette, E.G.: Performance comparison of laboratory and field produced pervious concrete mixtures. Constr. Build. Mater. **25**, 3187–3192 (2011). https://doi.org/10.1016/j.conbuildmat.2011.03.002
38. Sumanasooriya, M.S.; Neithalath, N.: Pore structure features of pervious concretes proportioned for desired porosities and their performance prediction. Cement Concr. Compos. **33**, 778–787 (2011). https://doi.org/10.1016/j.cemconcomp.2011.06.002
39. Asadi, S.; Hassan, M.M.; Kevern, J.T.; Rupnow, T.D.: Development of photocatalytic pervious concrete pavement for air and storm water improvements. Transp. Res. Rec. **2290**, 161–167 (2012). https://doi.org/10.3141/2290-21
40. Ibrahim, A.; Mahmoud, E.; Yamin, M.; Patibandla, V.C.: Experimental study on Portland cement pervious concrete mechanical and hydrological properties. Constr. Build. Mater. **50**, 524–529 (2014). https://doi.org/10.1016/j.conbuildmat.2013.09.022
41. Zhong, R.; Wille, K.: Compression response of normal and high strength pervious concrete. Constr. Build. Mater. **109**, 177–187 (2016). https://doi.org/10.1016/j.conbuildmat.2016.01.051
42. Elango, K.S.; Revathi, V.: Fal-G binder pervious concrete. Constr. Build. Mater. **140**, 91–99 (2017). https://doi.org/10.1016/j.conbuildmat.2017.02.086
43. Rezaei Lori, A.; Bayat, A.; Azimi, A.: Influence of the replacement of fine copper slag aggregate on physical properties and abrasion resistance of pervious concrete. Road Mater. Pavement Des. (2019). https://doi.org/10.1080/14680629.2019.1648311
44. Saboo, N.: Effect of fly ash and metakaolin on pervious concrete properties. Construction and Building Materials. 7 (2019)
45. Wang, H.; Li, H.; Liang, X.; Zhou, H.; Xie, N.; Dai, Z.: Investigation on the mechanical properties and environmental impacts of pervious concrete containing fly ash based on the cement-aggregate ratio. Constr. Build. Mater. **202**, 387–395 (2019). https://doi.org/10.1016/j.conbuildmat.2019.01.044
46. Yeih, W.; Chang, J.J.: The influences of cement type and curing condition on properties of pervious concrete made with electric arc furnace slag as aggregates. Constr. Build. Mater. **197**, 813–820 (2019). https://doi.org/10.1016/j.conbuildmat.2018.08.178
47. Zhou, H.; Li, H.; Abdelhady, A.; Liang, X.; Wang, H.; Yang, B.: Experimental investigation on the effect of pore characteristics on clogging risk of pervious concrete based on CT scanning. Constr. Build. Mater. **212**, 130–139 (2019). https://doi.org/10.1016/j.conbuildmat.2019.03.310
48. Gu, Y.; Zhang, D.; Bao, Z.: A new data-driven predictor, PSO-XGBoost, used for permeability of tight sandstone reservoirs: a case study of member of chang 4+5, western Jiyuan Oilfield, Ordos Basin. J. Petrol. Sci. Eng. **199**, 108350 (2021). https://doi.org/10.1016/j.petrol.2021.108350
49. Géron, A.: Hands-On Machine Learning with Scikit-Learn and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems. O'Reilly Media, Beijing ; Boston (2017)

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